From a failed move head to child-position value

fly_chess methods note

Martial Systems LLC. 2026-09-12. Fixture sha256
1924ff3b0c6e23b286ea33fd47357846ccb8095eb22cae790cf49278ba61c495.

Revisions

2026-09-12: first note. Records the closed hanging/teacher policy line, the decision to train child-position value, Stage A on mix-0 occupancy, and the mix-1 shuffle protocol.

2026-09-12: child-value table locked. A 0.91 vs random (n=200). B-real 0.4825, B-shuffle 0.5665. wiring_helped false. Mix-1 real hidden rates silent.

Abstract

A 1,007-cell fly-shaped leaky integrate-and-fire graph was given a chessboard as reserved-channel current and a legal-move mask on the way out. Two readouts were trained. The first asked whether real wiring makes a labeled legal move linearly easier than a degree-and-sign shuffle. It does not. Mix 0 is a typed occupancy register: a factored from-to head reads hanging (0.887) and a hanging-or-escape teacher (0.693) from that register. One mix ply removes those labels on real and shuffled wiring. After three plys the teacher delta includes 0. Hidden geometry still differs (knight-versus-empty cosine 0 vs 0.34 at mix 3). The wires do something. They do not pick the labeled move.

The failure mode is the training object, not the epoch count. The pipeline was: one position, write occupancy, run 1 to 3 LIF plys, linear or factored head votes a move. Mix 0 works because it is a 12 x 64 board register. Mix 1 and above fail because that register is gone, on real wiring and on shuffle. A hanging-heavy catalog of 130 puzzles, more epochs, or policy RL on those same mix-1 rates would turn a clean zero into a mushy almost. The sequel trains a scalar value of a child position. Mix 0 occupancy is the player. Mix 1 hidden rates, real versus five shuffle seeds, are the science. A two-layer MLP is the nonlinear probe.

1. Shared apparatus

MaleCNS v1.0 (Berg et al., *Cell* 2026) is a published adult male *Drosophila* central-nervous-system wiring diagram. Chess in this repo is not that table. Chess uses a 1,007-cell fixture (5,103 edges) with required roles resolved fail-closed, LIF mode fixture_voltage_jump (dt 0.5 ms, tau_m 20 ms, 40 steps per ply), a python-chess legal mask, and a degree-and-sign shuffle that permutes posts within each outgoing-sign class. Five shuffle seeds {1,2,3,4,5}. Graph frozen while a head trains. Chess play on the MaleCNS source is refused. The 475-cell MaleCNS slice is identity and circuit only: sugar lights MN9, loom lights DNp01, shuffle crosstalks. Named-cell Hz is the injected pulse. 8.0 current-units is a grid pick because 4.0 was silent.

2. Ethology paint (historical, n=40)

Hanging enemy pieces raise a sugar-like current. Check and hanging own pieces raise a looming-like current. Capture is MN9, flee is DNp01, quiet is BB/FG. Square identity is synthetic APP_LOCUS / AV_LOCUS. No linear head. Locked vs a uniform random legal mover, both colors, seed 0: score

0.4875 real and 0.4875 shuffled (40 games). Hanging-capture 0.177 vs 0.159 is not a win (774 vs 668 chances). Check-escape 1.0 is the mask after paint marks check (41 vs 31 positions). Leave that lock as history. It is not restamped onto the occupancy register or onto child-value.

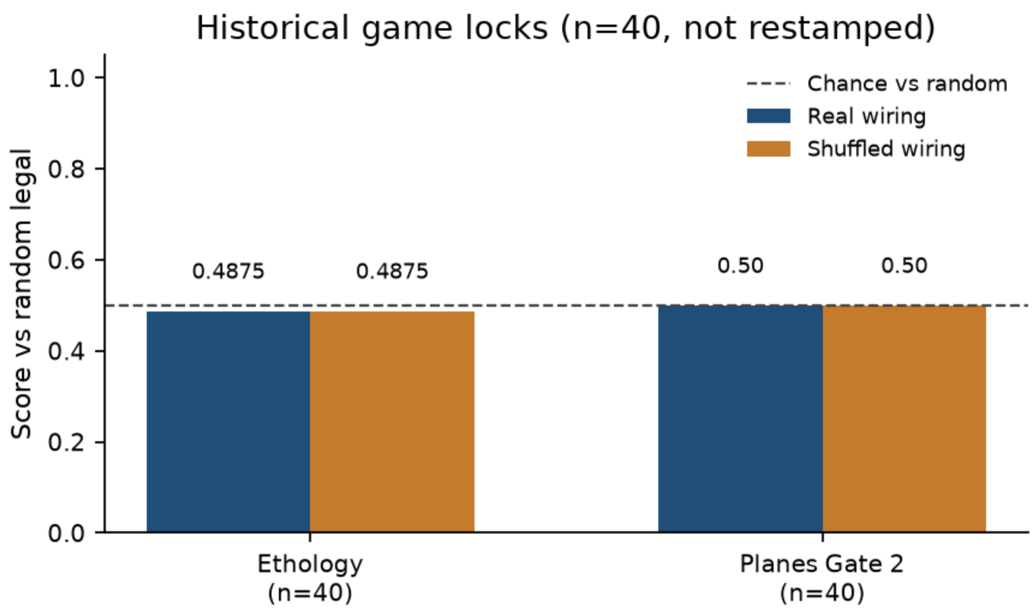

Figure 1. Historical game locks, n=40, random-legal opponent. Ethology 0.4875 / 0.4875. Planes Gate 2 0.50 / 0.50. Chance line is score vs random.

3. Planes policy (closed, 2026-09-10)

Twelve reserved pools, one cell per square per piece type, plus side-to-move, four castling cells, and eight en-passant file cells. Occupancy current 12.0. Mix 0 reads those reserved pools (synapses unused). Mix 1 and 3 run 1 or 3 LIF plys. Two heads: 4096-way from-to, and factored 64-from plus 64-to with the legal mask on the pair. Eight epochs, learning rate 0.08. Labels from injected board facts: hanging capture; check-escape (first legal escape by UCI sort); teacher = highest-value hanging capture, else that escape, else exclude. Split by FEN, no overlap. No engine.

Question: does the wiring make the labeled move linearly easier than shuffle? Figure of merit: accuracy real minus accuracy shuffle, same head, same FENs, same mix depth. Delta is allowed to be nonzero only when synapses are in the readout (mix 1 and above).

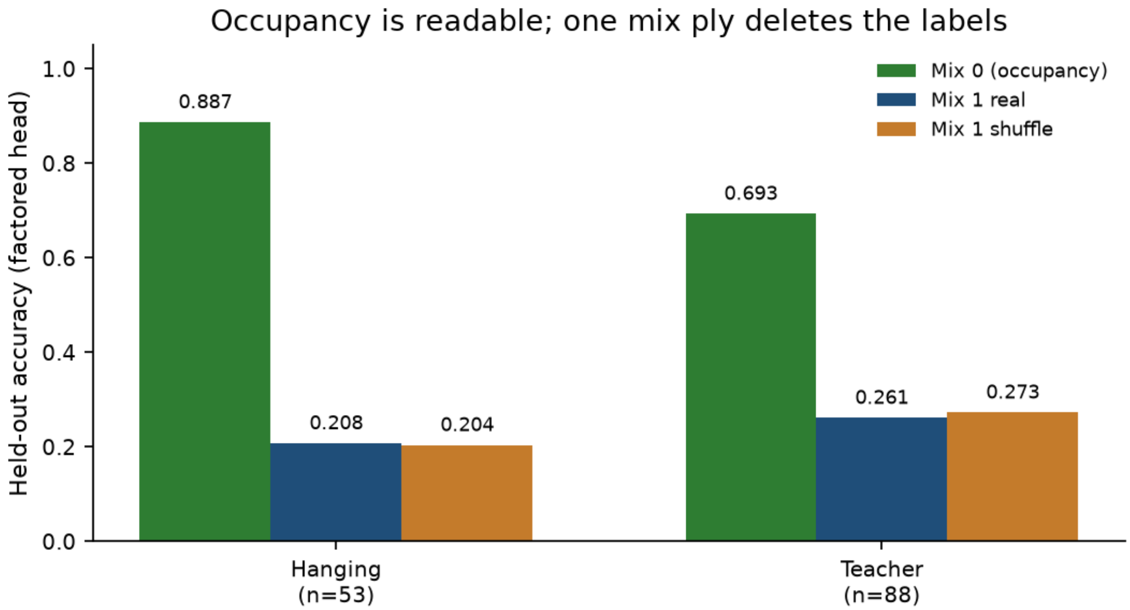

Figure 2. Occupancy is readable at mix 0. One mix ply deletes the hanging and teacher labels on both arms. Hanging n_eval=53, teacher n_eval=88, five seeds.

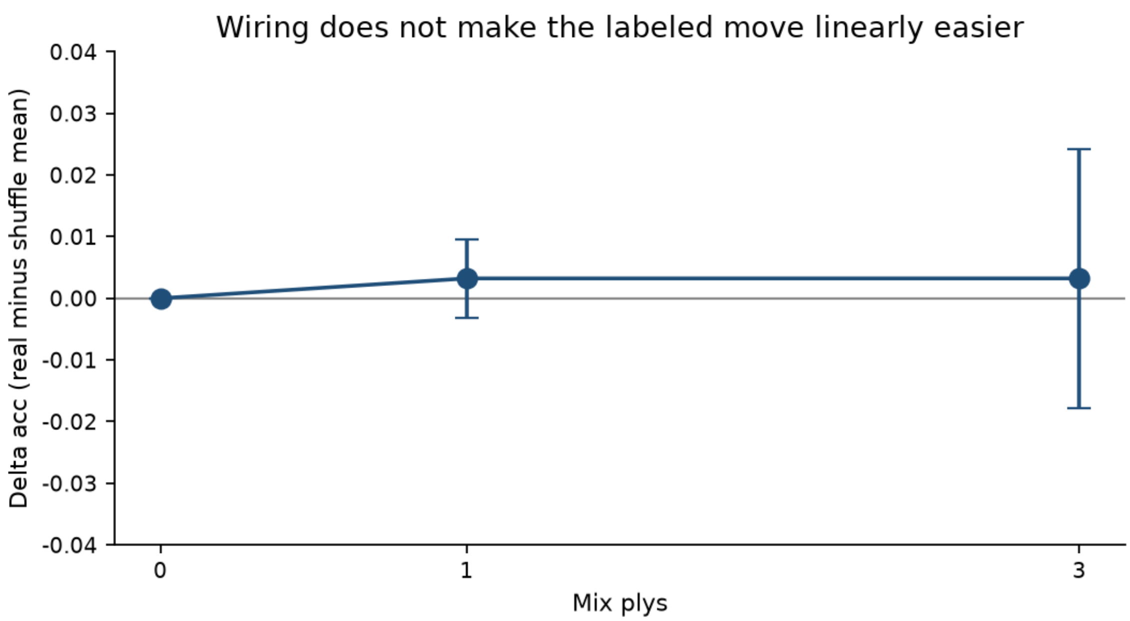

Figure 3. Shuffle-controlled delta across mix depth. Mix 3 teacher linear delta includes 0. Wiring does not make the labeled move linearly easier.

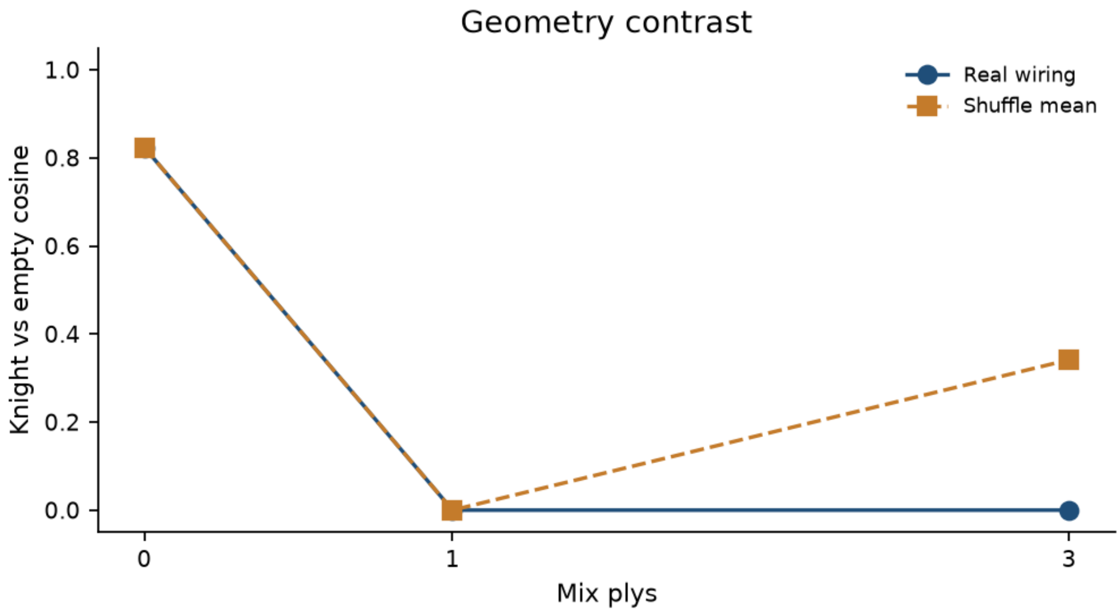

Figure 4. Knight-versus-empty cosine after three plys: 0 on real wiring, 0.341 on shuffle. Hidden geometry differs. The labeled move does not.

Locked label families (2026-09-10)

Family	Mix 0 factored real / shuffle	Mix 1 factored real / shuffle
Hanging	0.887 / 0.887	0.208 / 0.204
Teacher	0.693 / 0.693	0.261 / 0.273

Check-escape flee_ok = 1.0 at every depth on both arms: the legal mask is the decoder. Do not quote check-escape exact-match as occupancy or wiring.

Locked mix-depth, 4096-way head, 130 train / 62 eval (2026-09-10)

Mix plys	Real acc	Shuffle acc	delta acc mean [95%]	Knight-empty cosine
0	0.226	0.226	0.00 [0.00, 0.00]	0.824 / 0.824
1	0.177	0.174	0.003 [-0.003, 0.010]	0.000 / 0.000
3	0.177	0.174	0.003 [-0.018, 0.024]	0.000 / 0.341

Factored occupancy read at mix 0 is 0.790, delta 0. Mix hurts both arms. The reservoir does not refine occupancy into tactics. It throws occupancy away.

4. What the policy loop actually trained (2026-09-12)

The closed line is a linear classifier on a labeled legal move. At mix 0 the features are injection currents on the occupancy register. A factored head can read hanging and teacher from that register because those labels are board facts still sitting on typed cells. After one LIF ply the same head is reading mixed rates. On this fixture the 64 HIDDEN cells stay subthreshold on real wiring (start-position hidden rates all 0 Hz; voltage rises from -52 mV toward -45 mV and does not spike). Shuffle rewiring lets a few hidden cells spike. Cosine therefore splits the arms. Linear move-class does not.

The catalog made the mix-0 head look smart and every mixed readout look broken for a second reason. It is hanging-heavy, teacher-sparse, and it drops quiet positions. A 4096-way policy head trained on that set is a hanging detector on a register, then noise after a ply. More FENs of the same kind, more epochs at mix 3, or RL on those mix-1 rates, keep the same object. The clean zero would become an interval that includes 0 with a story attached.

Enumerating legal moves, scoring child boards, and picking the best is a 1-ply evaluator. That loop uses the occupancy layout. It is allowed as a ranker. It is not a claim that the fly produced the move as a policy from mixed rates.

5. Child-position value (open, 2026-09-12)

Question: On dense game positions, does a mix-0 occupancy value head that scores legal children beat random-legal, and do mix-1 hidden rates on real wiring beat a degree-and-sign shuffle at the same value task?

A. Mix-0 value head (the player). Input: 12 x 64 occupancy planes plus side-to-move, castling, and en passant (781 currents, synapses unused). Output: a scalar, White-centric pawn units, material plus a small piece-square table. At test: for every legal move, push, score the child, pick the max for the mover. Ridge on parents and all legal children of the train games.

B. Mix-1 value head (the science). Same targets. Features: 64 HIDDEN-cell rates after one ply. One copy on real wiring, one on each of five shuffle seeds. If B-real matches B-shuffle, the synapses still are not playing. If B-real beats B-shuffle by a real delta on held-out value and on games, there is a wiring effect. Cosine already says the dynamics differ. Cosine does not say the difference is chess.

C. Two-layer MLP on mix-1 rates. Geometry can differ and still be linearly useless. Hidden width 32, tanh, same train and eval FENs, same shuffle seeds. If the MLP delta includes 0, that line stops too.

Data: 320 self-play games from the start position under random, capture-preferring, and 1-ply material policies. Sixteen positions per game. Hold out by game_id, not by row. Train 4,096 positions (256 games), eval 1,024 (64 games). Phase mix on train: opening 1,068, middlegame 1,998, simple 1,030. Quiet positions stay in. Every legal child of an eval parent is scored. Opponent at play is random legal. Report score vs that opponent and vs the mix-0 head. Ethology sugar/loom stays a separate arm (locked 0.4875). It is not the main trainer.

6. Order of operations

1. Get A playing. If mix-0 child scoring cannot beat random by a lot, the head or the data are wrong and the fly is irrelevant.

2. Freeze A. Train B and C on the same targets.

3. Table three numbers on the same eval set: A, B-real, B-shuffle. Games n at least 200, not n=40.

4. Only if B-real beats both A and B-shuffle on held-out value and on games is wiring_helped true.

5. Only then unfreeze anything. Unfreezing synapses first is a sparse RNN that was initialized as a fly. Valid experiment, different claim: fly init versus shuffle init versus random sparse init. Publish that as

init.

Kept off this line: more epochs on hanging labels at mix 3; the 4096-way head as the player; a ply loop on 166k MaleCNS cells; tuning until shuffle almost loses, then dropping the shuffle; policy RL before the value-on-children table.

7. Stage A mix-0 ranker (2026-09-12)

Ridge on occupancy currents recovers the hand eval (Pearson 0.9997 on 1,024 held-out positions, MSE 0.0265). Child pairwise ranking vs the same target is 0.959; best-child match 0.798 (n_ranked=1022). Versus random legal, n=200, both colors, seed 0: score 0.91 [0.870, 0.950]. Illegal moves 0. beats_random is true. Synapses unused. Elo is null. Call it a ranker sitting on a fly-shaped input layout. The ceiling exists. The fly is now relevant as a mix-1 probe, not as this player.

Arm	Features	Held-out pairwise	Games n=200 vs random
A mix-0	occupancy 12 x 64 + STM/castle/EP	0.959	0.91 [0.870, 0.950]
B-real mix-1	HIDDEN 64 after one ply, real wiring	0.007	0.4825
B-shuffle mix-1	same features, seeds 1 to 5	0.265	0.5665

B-real hidden rates are silent (train_rate_max 0.0 Hz). Child pairwise 0.007; games 0.4825. Shuffle spikes a few hidden cells: pairwise 0.265, games 0.5665 (five seeds, n=200 each). Games delta real minus shuffle -0.084 [-0.110, -0.058]. wiring_helped is false. Mix-0 occupancy ranker is the player. Mix-1 real did not beat both mix-0 and shuffle, so the synapses still are not playing.

C, two-layer MLP on the same mix-1 rates: real pairwise 0.007, shuffle pairwise 0.260. Pairwise delta -0.253 [-0.266, -0.239]. The interval excludes 0 and the sign is negative: real is silent, shuffle leaks spikes. That is not a wiring win.

8. What holds (2026-09-12)

The mix-0 occupancy ranker is the player. Mix-1 real hidden rates do not spike, so linear and MLP value heads on those rates cannot rank chess. Shuffle wiring produces a few hidden spikes and a weak ranker that still loses to mix-0 by a wide margin. wiring_helped is false. unfreeze_allowed is false. The mix-1 value line stops. Cosine remains a geometry fact. Ethology 0.4875 and Gate 2 n=40 stay historical. Do not drop a shuffle seed. Do not grind hanging labels at mix 3.

A later arm, if any, is init: train sparse RNN weights from fly wiring versus shuffle versus random sparse, and say so. Self-play reward on the mix-0 ranker, if we get that far: +1 checkmate, 0 draw, -1 loss, a small term for material after a few plies. Ethology sugar/loom stays a separate arm. It already scored 0.4875.

9. Files

Path	Role
docs/planes_note.md	Closed hanging/teacher policy write-up
docs/value_note.md	Child-value methods
logs/planes_labels.json	Label-family mix table
logs/planes_mix.json	Mix-depth by five seeds

Path	Role
logs/ethology_gate.json	Paint controller, n=40
logs/value_lock.json	Child-value table
config/value.json	Split, freeze, shuffle seeds
src/fly_chess/train_value.py	A, B, C
data/fixtures/graph.json	1,007-cell fixture

Reproduce with .venv/bin/python -m pytest. Rebuild this PDF with .venv/bin/python scripts/make_method_note.py after pip install -e ".[figures]".

Generated 2026-09-12. Build clock. It does not replace the revision dates above.